Recognizable extensions of colored graphs

Human Compression Project

Abstract

A construction can carry every symmetry of its input and still acquire additional symmetries. We separate these two questions by a restriction and extension criterion for permutation groups and their distinguished subgroups. Intrinsic recognition of retained vertices, rigidity over those vertices, and explicit extension identify the full group. Component boundaries provide a useful local rigidity test. We give complete proofs for the chamber completion and cleaning constructions of Díaz Ramos, Molinier and Viruel. The application paper uses these results to recover their finite normal-pair representation theorem and recursive symmetric-group family.

1 Restriction, extension and component boundaries

An incidence system in this paper is a nonempty finite simple graph X, together with a surjective proper coloring $t : V(X) \to I$. Thus adjacent vertices have different colors. A *correlation* is a graph automorphism f that permutes the color classes: there is a permutation σ_f of I such that $t(f(v)) = \sigma_f(t(v))$ for all v. Surjectivity makes σ_f unique. Write $C(X)$ for the correlation group and $K(X)$ for its subgroup preserving each color. The map $f \mapsto \sigma_f$ is a homomorphism, so $K(X)$ is normal and $C(X)/K(X)$ embeds in $\mathrm{Sym}(I)$. In general $C(X)$ need not be the full uncolored graph automorphism group.

A *chamber* is a clique meeting every color class. An incidence system is an *incidence geometry* when each clique, including the empty clique, is contained in a chamber. No transitivity, thinness or residual connectedness is part of this definition. These are the conventions of Díaz Ramos–Molinier–Viruel [1], Definitions 2.1–2.3.

The following elementary fact is the common group-theoretic argument. It is stated for arbitrary permutation groups; finiteness enters the graph constructions later.

Theorem 1 (Restriction and extension)**.** *Let $\iota : V \hookrightarrow W$ be an injection, let $A \le \mathrm{Sym}(V)$ and $B \le \mathrm{Sym}(W)$, and let $A_0 \le A$ and $B_0 \le B$. Suppose that:*

1. *for each $b \in B$ there is $a \in A$ such that $b\iota = \iota a$;*
2. *a member of B fixing $\iota(V)$ pointwise is the identity;*
3. *every $a \in A$ has an extension $b \in B$ satisfying $b\iota = \iota a$;*
4. *for such a pair (b, a), one has $b \in B_0$ if and only if $a \in A_0$.*

Then restriction induces an isomorphism $B \cong A$ carrying B_0 onto A_0. Every extension in the third condition is unique, and the resulting extension map is the inverse homomorphism.

Proof. The injection ι makes the element a in the first condition unique; write $\rho(b) = a$. For $b, c \in B$,

$$bc\iota = b\iota\rho(c) = \iota\rho(b)\rho(c),$$

so uniqueness gives $\rho(bc) = \rho(b)\rho(c)$. The same argument gives $\rho(1) = 1$. Its kernel consists exactly of the elements fixing $\iota(V)$ pointwise, and is trivial by the second condition. The third condition makes ρ surjective. The last condition gives $\rho(B_0) = A_0$: containment follows immediately, while a preimage of any $a \in A_0$ belongs to B_0. Thus ρ is the asserted pair isomorphism. Injectivity gives uniqueness of extensions, and the inverse of an isomorphism is a homomorphism. □

For colored graphs X, Y, one applies this theorem to $A = C(X)$, $B = C(Y)$ and their palette kernels. Recognizing a retained copy of $V(X)$ is not enough: the restricted permutation must preserve the *reconstructed* old incidences and colors. In particular, old edges need not remain edges of Y. Conversely, an explicit action of $C(X)$ on Y establishes only the extension condition, until rigidity rules out further correlations.

Here is a local criterion for that rigidity. All induced graphs and boundaries in its statement are taken in the constructed graph Y.

Lemma 2 (Recognition by component boundaries)**.** *Let $S \subseteq V(Y)$ be invariant under a group B of correlations of Y. For each connected component D of $Y \setminus S$, define its boundary*

$$\partial D = \{s \in S : s \text{ has a neighbor in } D\}.$$

Assume distinct components have distinct boundaries. Suppose also that any correlation of the induced colored graph on $D \cup \partial D$ fixing ∂D pointwise fixes D pointwise. Then a member of B fixing S pointwise is the identity on Y.

Proof. Such a member permutes the components of $Y \setminus S$ and sends each boundary to itself. Distinctness of the boundary sets therefore fixes each component setwise. Its restriction to $D \cup \partial D$ is a correlation, fixes the boundary pointwise, and is the identity by the local assumption. This holds for every component; it is already the identity on S. □

Only color classes occurring in an induced subgraph are used when speaking of its restricted coloring. A global correlation preserving that subgraph does permute those classes, even if it permutes other colors outside the subgraph. Thus the local hypothesis in Lemma 2 does not silently replace correlations by color-preserving automorphisms.

2 Completing edges to chambers

The following construction and theorem are due to Díaz Ramos, Molinier and Viruel [1, Theorem 4.1]. We retain their construction and prove all four conditions of Theorem 1.

Theorem 3 (Chamber completion)**.** *Let X be a finite properly colored triangle-free graph of minimum degree at least two, with palette I of size r. For every edge $e = \{v, w\}$ and every $i \in I \setminus \{t(v), t(w)\}$, introduce a vertex $u_{e,i}$ of color i. Make*

$$B_e = \{v, w\} \cup \{u_{e,i} : i \notin \{t(v), t(w)\}\}$$

a complete graph, retaining all old vertices and adding no other edges. The resulting graph $J(X)$ is an incidence geometry on the same palette and restriction gives

$$(C(J(X)), K(J(X))) \cong (C(X), K(X)).$$

It has $|V(X)| + (r-2)|E(X)|$ vertices and the same connected components as X.

Proof. Every B_e has exactly one vertex of each color. A clique containing a new vertex lies entirely in its B_e, since that vertex has no neighbors outside B_e. A clique consisting of old vertices has size at most two by triangle-freeness. A two-vertex clique supplies its edge e; a singleton extends using an incident edge; the empty clique extends because X has an edge. Hence every clique is contained in a chamber.

A new vertex has degree $r - 1$. At an old vertex v, the sets $B_e \setminus \{v\}$, for edges e through v, are pairwise disjoint. Consequently

$$\deg_{J(X)}(v) = \deg_X(v)(r-1). \tag{1}$$

The hypotheses imply $r \geq 2$ and $\deg_X(v) \geq 2$, so the old vertices are exactly those of degree greater than $r-1$. Every correlation recognizes them. Their induced graph is X, and their old colors are unchanged; the restriction therefore belongs to $C(X)$.

Suppose a correlation fixes every old vertex. Every old color occurs, so it fixes every color. If a new vertex has old neighbor set $\{v, w\}$ and color i, it must be $u_{\{v,w\},i}$. Thus all new vertices are fixed and restriction is injective. If $f \in C(X)$ induces $\sigma_f \in \mathrm{Sym}(I)$, define

$$\widetilde{f}(v) = f(v), \qquad \widetilde{f}(u_{e,i}) = u_{f(e),\sigma_f(i)}.$$

This is a bijection, with inverse defined from f^{-1}. It carries each B_e onto $B_{f(e)}$, and hence preserves adjacency in both directions. It fixes all colors precisely when f does. Theorem 1 now identifies the pairs. When $r = 2$ no vertices are added, and this proof still applies.

There are $r-2$ new vertices per edge, giving the count. Every new vertex is attached within the component of its edge, and all old edges are retained. No components are joined or lost. □

The source's degree argument on physical page 7 uses r in place of $r-1$. Equation (1) is the actual degree calculation. The strict separation needed for its theorem is unaffected. The statement concerns this construction and these hypotheses; it does not assert that they are logically necessary for every conceivable pair-preserving completion.

3 Cleaning arbitrary incidence systems

The next construction is [1, Theorem 4.2]. We include the rigidity of its tail and the isolated-vertex case, since those are the local obligations that make its symmetry assertion valid.

Theorem 4 (Cleaning)**.** *Every nonempty finite properly colored graph X has a finite properly colored, triangle-free graph $L(X)$ of minimum degree at least two satisfying*

$$(C(L(X)), K(L(X))) \cong (C(X), K(X)).$$

It preserves the connected-component decomposition and adds exactly $q(2M+5)$ vertices, where

$$M = \max_{i \in I} |t^{-1}(i)|, \qquad W = \{v : \deg_X(v) \leq 1\}, \qquad q = |E(X)| + |W|.$$

Its palette consists of the old colors and two new colors.

Proof. Since X is nonempty, $M \geq 1$; also $q \geq 1$, since otherwise a nonempty edgeless graph would have no vertices of degree at most one. For each edge e and each $w \in W$ take a separate gadget with vertices $u_0, \ldots, u_{2M+4}$, consecutive edges $u_j u_{j+1}$, and the two chords

$$u_{2M+4}u_{2M-1}, \qquad u_{2M+3}u_{2M}. \tag{2}$$

Delete all old edges. For $e = \{v, w\}$ join both endpoints to the root $u_{e,0}$ of its edge gadget. Join each $w \in W$ to $u_{w,0}$; if w was isolated, also join it to $u_{w,2}$. Retain old colors, color even gadget indices by α and odd indices by β, where α, β are distinct new colors.

Every new edge joins different colors. An old vertex of degree at least two retains that degree; a former leaf acquires a second neighbor; a former isolated vertex acquires its two displayed neighbors. Every gadget vertex has degree at least two. There are no old–old edges. The new induced graph is bipartite. A triangle containing an old vertex would require two adjacent new neighbors in one gadget, but the only old vertex with two neighbors in one gadget was isolated, and its neighbors u_0, u_2 are not adjacent. Thus $L(X)$ is triangle-free and has the required minimum degree.

The two new color classes have sizes $q(M+3)$ and $q(M+2)$. These sizes are distinct and both exceed M, the largest old class size. Every correlation therefore fixes each of these classes and preserves

the old vertex set. Two distinct old vertices were adjacent in X exactly when they have a common new neighbor in $L(X)$: their edge gadget supplies its root, and every other gadget has only one old neighbor. This reconstructs the old graph as well as its color partition, so restriction belongs to $C(X)$.

To establish rigidity, delete all old vertices. The components are precisely the separate gadgets. Each edge gadget has its old endpoint pair as boundary; each additional gadget has its one old vertex as boundary. The graph is simple and at most one additional gadget is attached to a given old vertex, so all these boundaries are distinct.

Consider a correlation of a gadget together with its boundary that fixes the boundary. For an edge gadget or a former leaf, the root is the unique gadget vertex adjacent to the boundary. For an isolated vertex the two attached vertices u_0, u_2 have different degrees in this induced graph: the first has degree two and the second at least three. Hence the root is fixed in all cases. Starting at the root forces successive vertices until u_{2M-1}. At that vertex the forward neighbors u_{2M} and u_{2M+4} have induced-gadget degrees three and two respectively, so both are fixed. At u_{2M} the remaining neighbors u_{2M+1} and u_{2M+3} have induced-gadget degrees two and three respectively. Finally u_{2M+2} is forced. These branch degrees are taken within the gadget, after its component has been recognized; an isolated attachment therefore causes no exception when $M = 1$. Lemma 2 now gives pointwise rigidity over all old vertices.

A correlation f of X preserves degrees. Extend it by

$$u_{e,j} \longmapsto u_{f(e),j}, \qquad u_{w,j} \longmapsto u_{f(w),j}.$$

The complete edge list, including the isolated attachment, is preserved, and the inverse is the same construction from f^{-1}. Both new colors are fixed; old colors are fixed exactly when f fixes them. Thus Theorem 1 gives the asserted pair isomorphism. There are q gadgets of $2M + 5$ vertices. Replacing an old edge preserves connectivity between its endpoints and does not join different old components; all other gadgets attach to one old vertex. This proves the component claim. □

Corollary 5. *Every finite incidence system has a finite incidence geometry with the same correlation group and the same distinguished type-preserving subgroup. Connected inputs give connected outputs.*

Proof. Apply Theorem 3 to $L(X)$ from Theorem 4. Both restriction maps identify the distinguished kernels. Each construction preserves the component decomposition. □

4 Attribution and verification

The two constructions, their symmetry assertions and their use in the finite normal-pair theorem are due to Díaz Ramos, Molinier and Viruel. The restriction-and-extension criterion is elementary classical reasoning; its explicit separation from recognition and rigidity is an organization of the proofs, not a priority claim. Lemma 2 isolates the component argument also used in the recursive application. No external graph representation theorem is needed to make the proofs above complete.

The accompanying `RecognizableRestriction.lean` constructs the restriction homomorphism from its actual action on an embedded vertex set, proves its kernel description, derives injectivity from pointwise rigidity and surjectivity from existence of extensions, constructs the group equivalence, and transports a distinguished subgroup. It does not formalize the incidence or degree arguments proving those hypotheses for any gadget. The complete chamber, cleaning and component arguments above remain written proofs.

The two complete GAP programs in the separate appendix provide bounded checks, including exceptional graphs and small normal pairs. Their computations do not replace any quantified theorem above. GAP and GRAPE's exact finite automorphism algorithms remain part of the trust boundary of those checks; Lean uses its kernel and the pinned mathlib infrastructure. Neither the number of programs nor the number of formal declarations measures mathematical compression.

References

[1] A. Díaz Ramos, R. Molinier and A. Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025.